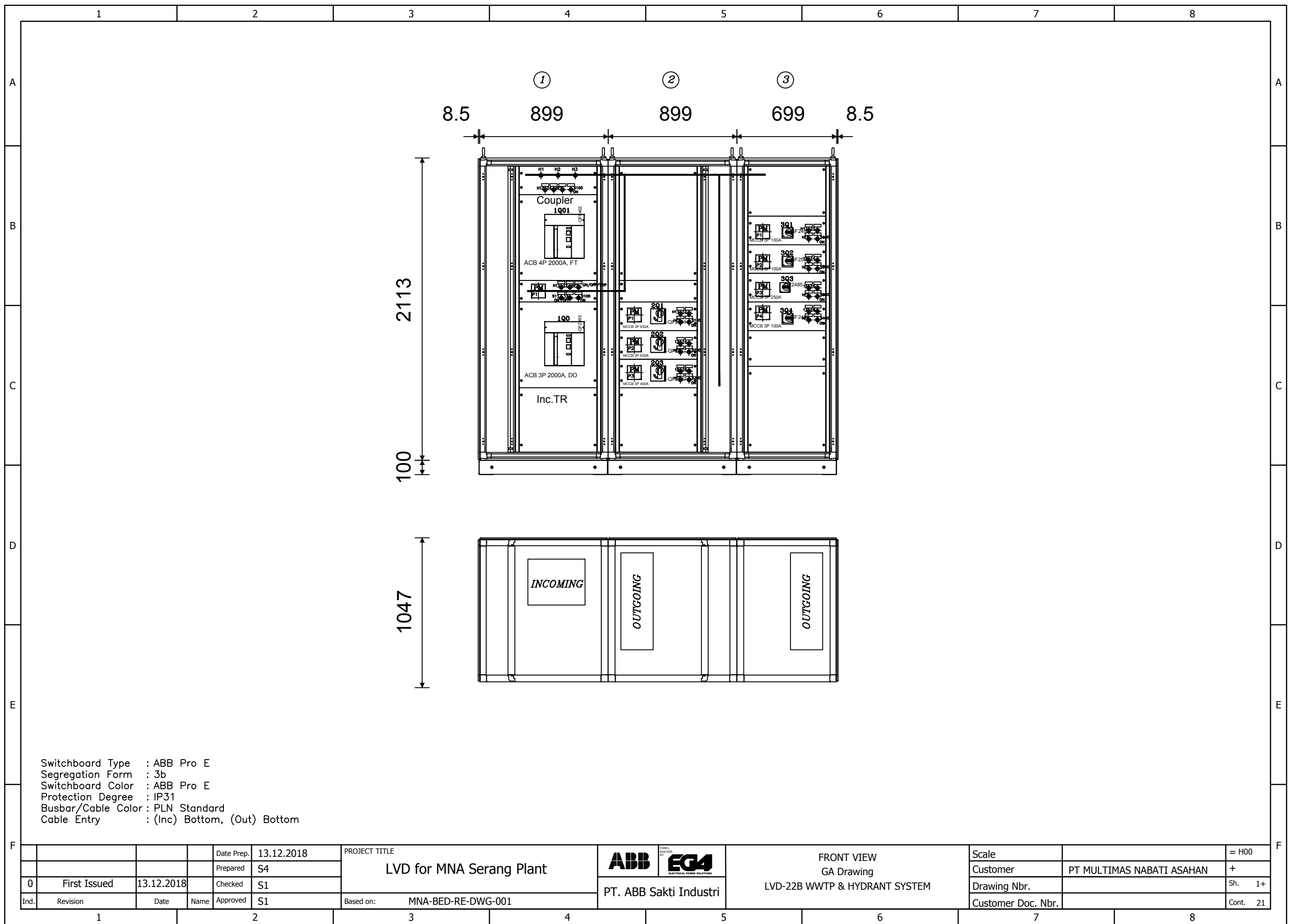
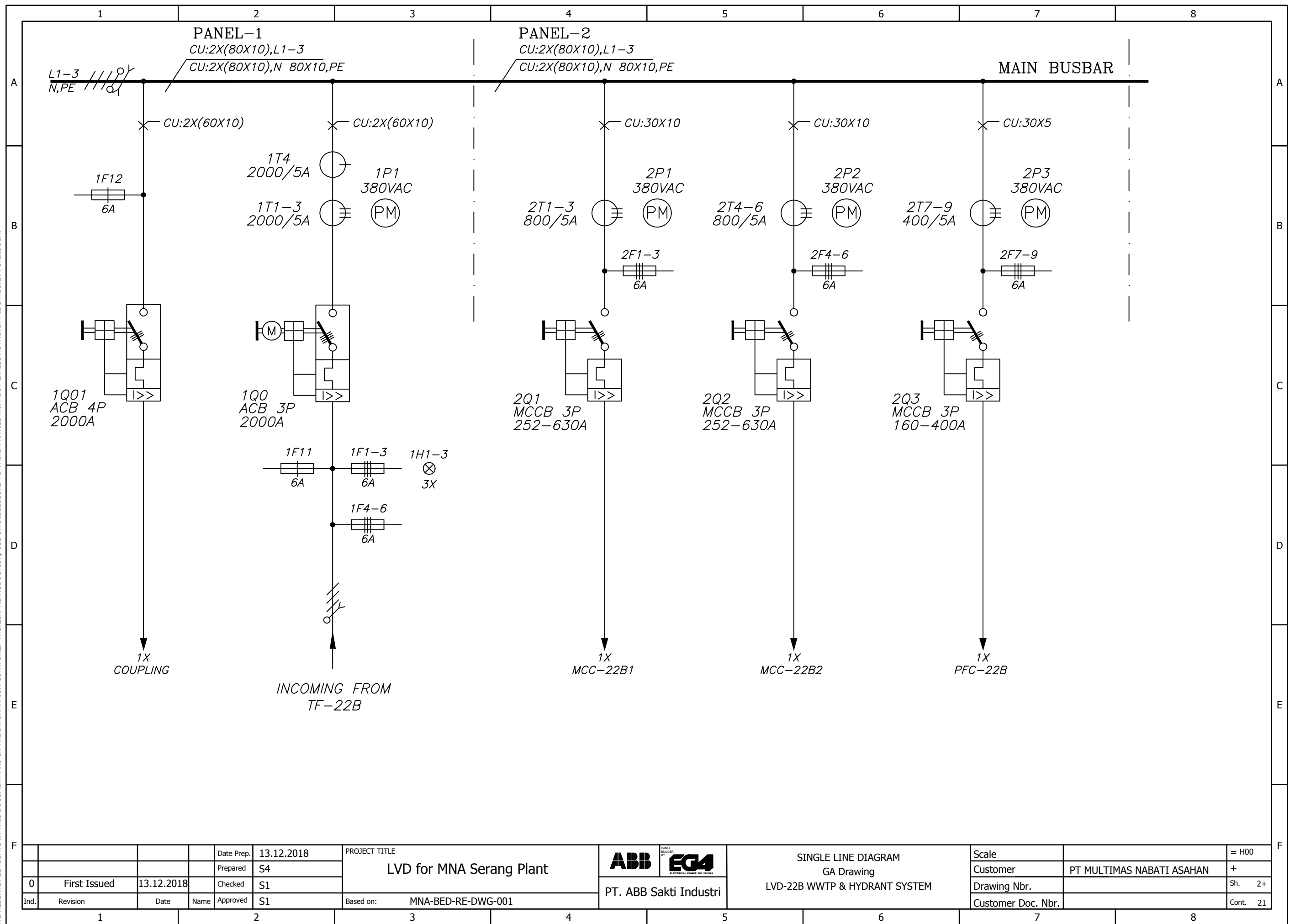


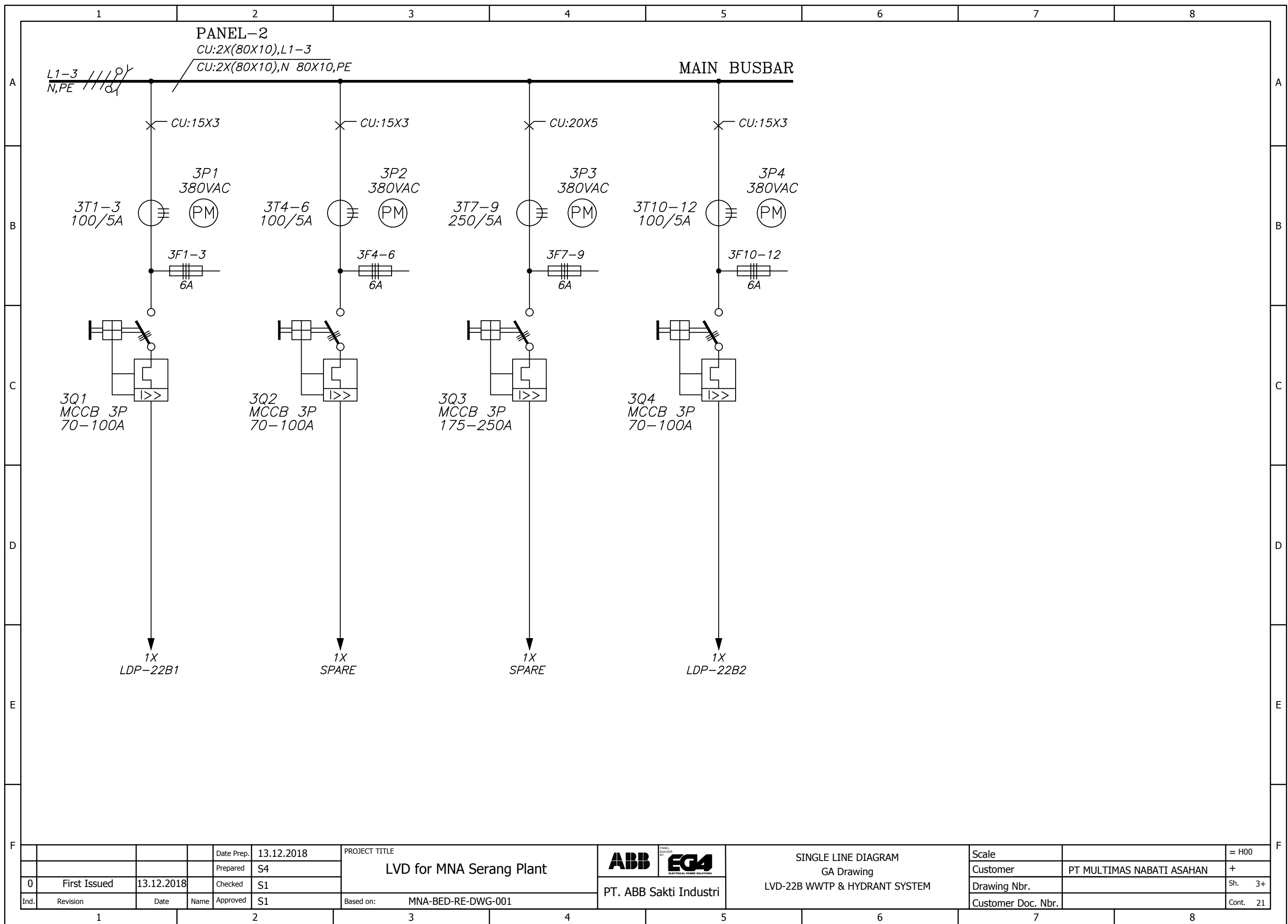
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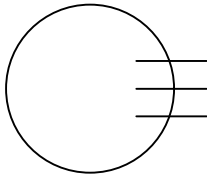
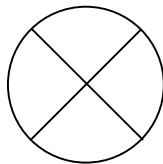
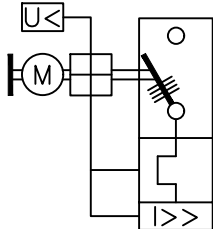
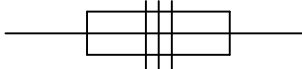
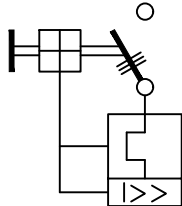
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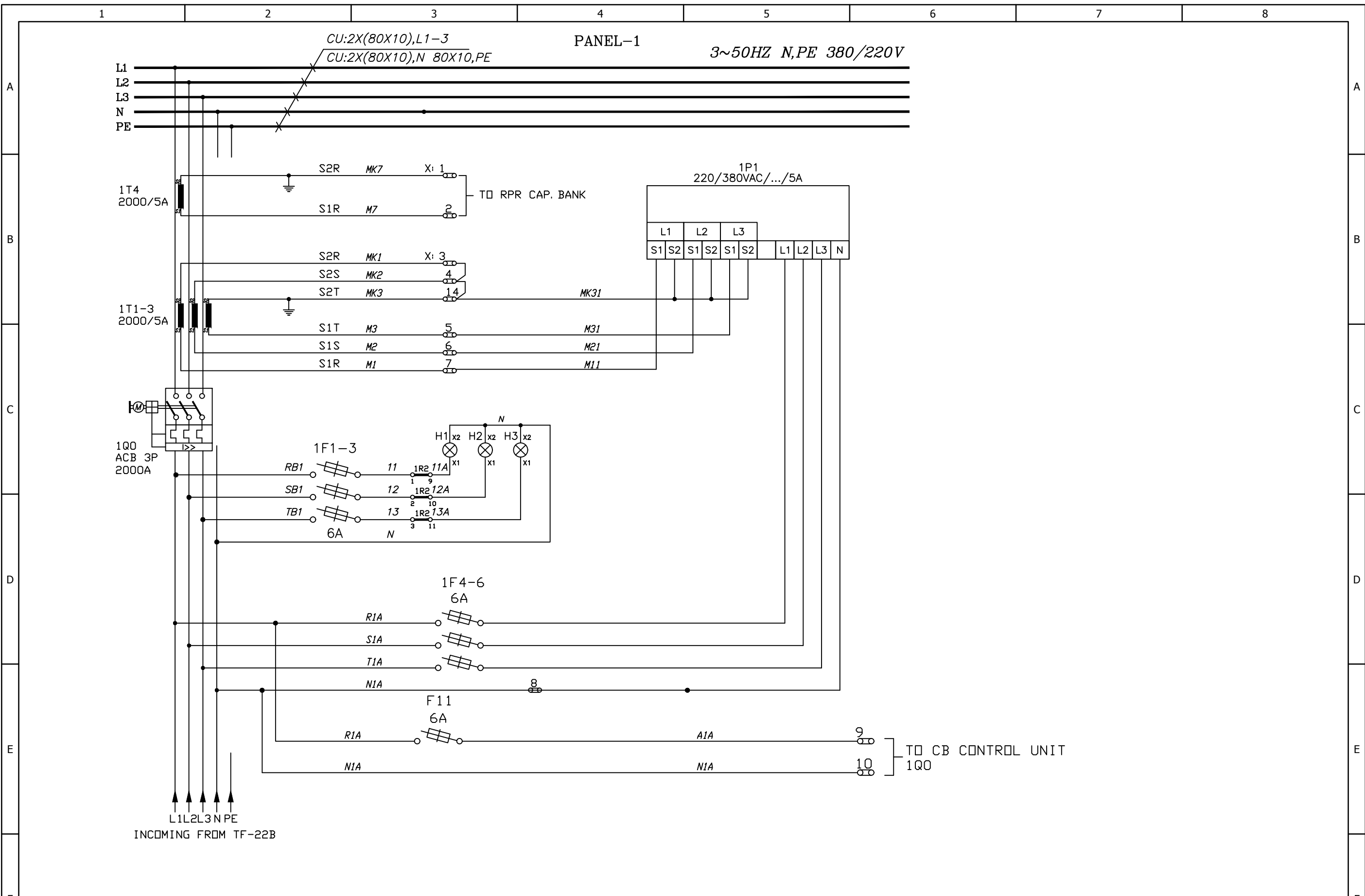
1		2		3		4		5		6		7		8			
A																A	
				CT						PILOT LAMP							
B																B	
				ACB						FUSE							
C																C	
				MCCB													
D																D	
				DIGITAL POWER METER													
E																E	
F																F	
				Date Prep.	13.12.2018	PROJECT TITLE LVD for MNA Serang Plant			LEGEND GA Drawing LVD-22B WWTP & HYDRANT SYSTEM			Scale		= H00			
				Prepared	S4							Customer	PT MULTIMAS NABATI ASAHAN		+		
0	First Issued	13.12.2018		Checked	S1							Drawing Nbr.		Sh.	4+		
Ind.	Revision	Date	Name	Approved	S1	Based on: MNA-BED-RE-DWG-001						Customer Doc. Nbr.		Cont.	21		
1		2		3		4		5		6		7		8			

ABB

PANEL
DESIGNER
BY
EG4
ELECTRICAL POWER SOLUTIONS

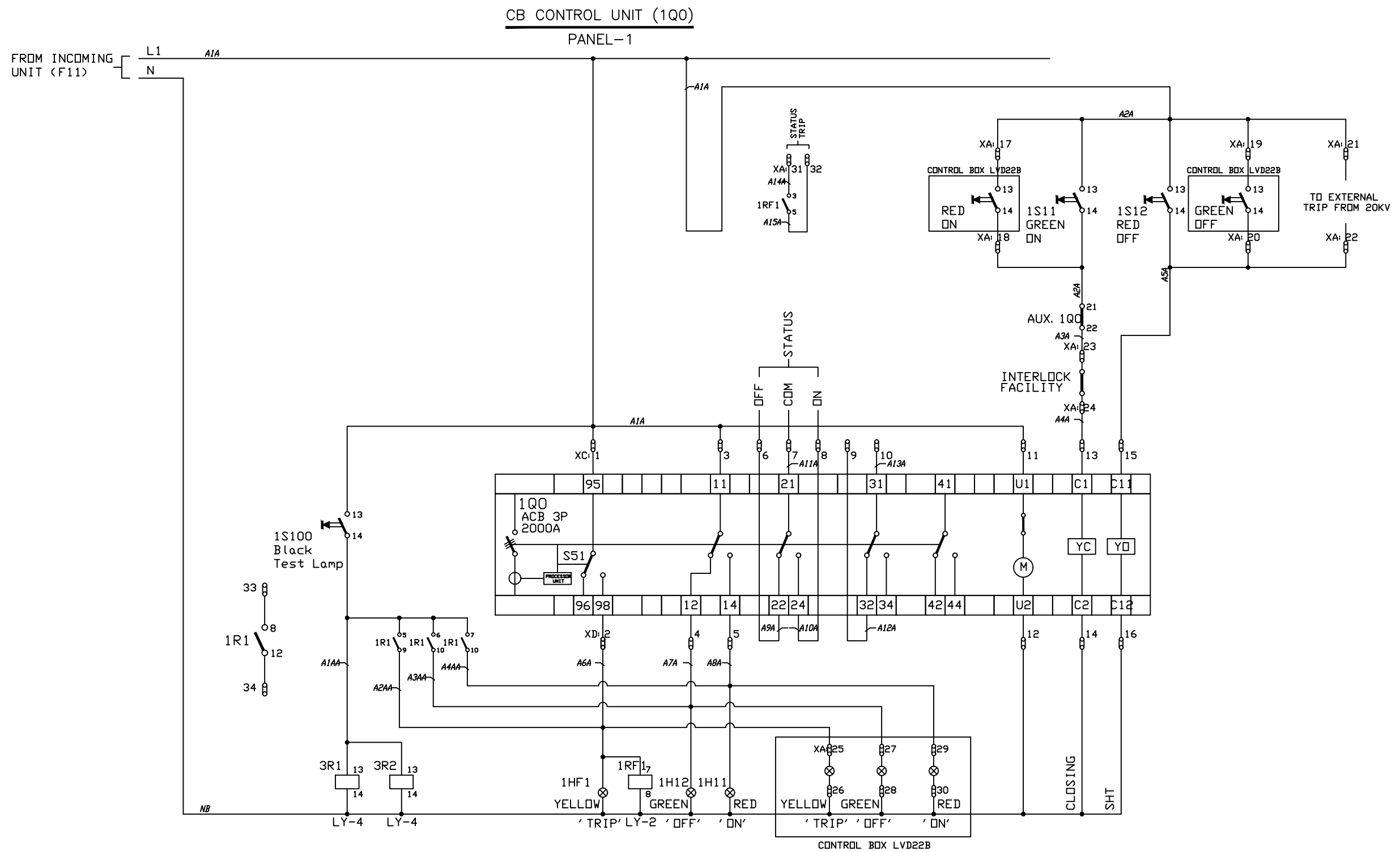
PT. ABB Sakti Industri

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		Date Prep.	13.12.2018	PROJECT TITLE LVD for MNA Serang Plant		ABB PANEL SELECTOR BY EG4 PT. ABB Sakti Industri	WIRING CONTROL DIAGRAM GA Drawing LVD-22B WWTP & HYDRANT SYSTEM	Scale		= H00
		Prepared	S4					Customer	PT MULTIMAS NABATI ASAHAN	+
0		First Issued	13.12.2018					Drawing Nbr.		Sh. 1
Ind.	Revision	Date	Name	Approved	S1	Based on:	MNA-BED-RE-DWG-001	Customer Doc. Nbr.		Cont. 21

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		Date Prep.	13.12.2018	PROJECT TITLE LVD for MNA Serang Plant		ABB EG4 ELECTRICAL POWER SOLUTIONS		WIRING CONTROL DIAGRAM GA Drawing LVD-22B WWTP & HYDRANT SYSTEM		Scale	= H00	
		Prepared	S4							Customer	PT MULTIMAS NABATI ASAHAN	
0		First Issued	13.12.2018							Drawing Nbr.	Sh. 1	
Ind.	Revision	Date	Name	Approved	S1	Based on:	MNA-BED-RE-DWG-001		PT. ABB Sakti Industri		Customer Doc. Nbr.	Cont. 21

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				Date Prep.	13.12.2018	PROJECT TITLE LVD for MNA Serang Plant	ABB PANEL BUILDER EG4 ELECTRICAL POWER SOLUTIONS	WIRING CONTROL DIAGRAM GA Drawing LVD-22B WWTP & HYDRANT SYSTEM	Scale		= H00
				Prepared	S4				Customer	PT MULTIMAS NABATI ASAHAN	+
0	First Issued	13.12.2018		Checked	S1				Drawing Nbr.		Sh. 1
Ind.	Revision	Date	Name	Approved	S1				Based on:	MNA-BED-RE-DWG-001	Customer Doc. Nbr.
1	2	3	4	5	6	7	8				

PANEL-1

CU:2X(80X10),L1-3
CU:2X(80X10),N 80X10,PE

L1
L2
L3
N
PE

1F12
6A
RIA 1A

Aux TRIP 1Q01
Aux ON/OFF 1Q01

1R3 1R3 1R3 1R3

1S200 Black Test Lamp

1HF2 YELLOW
1H13 GREEN
1H13 RED

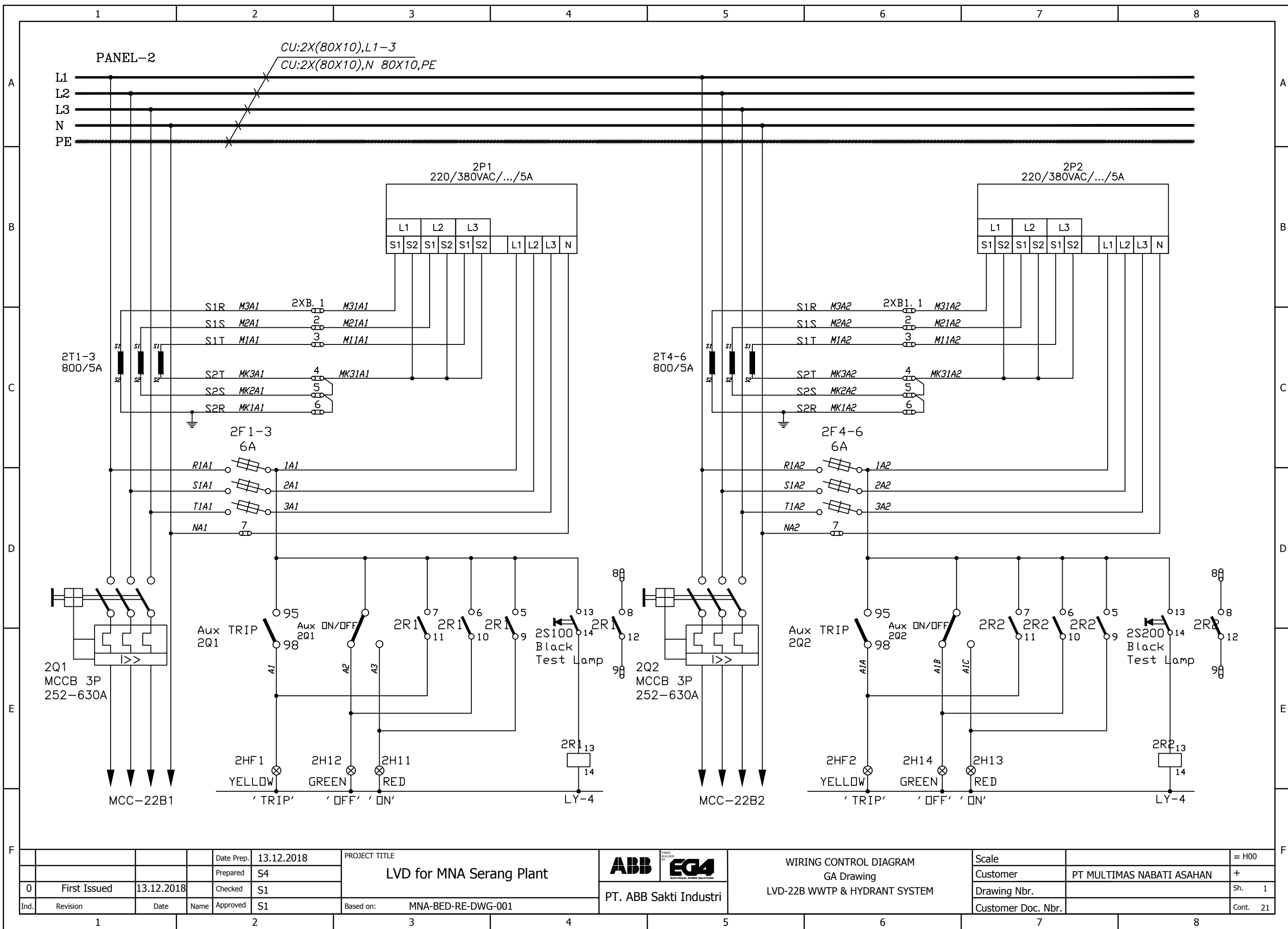
'TRIP' 'OFF' 'ON'

LY-4

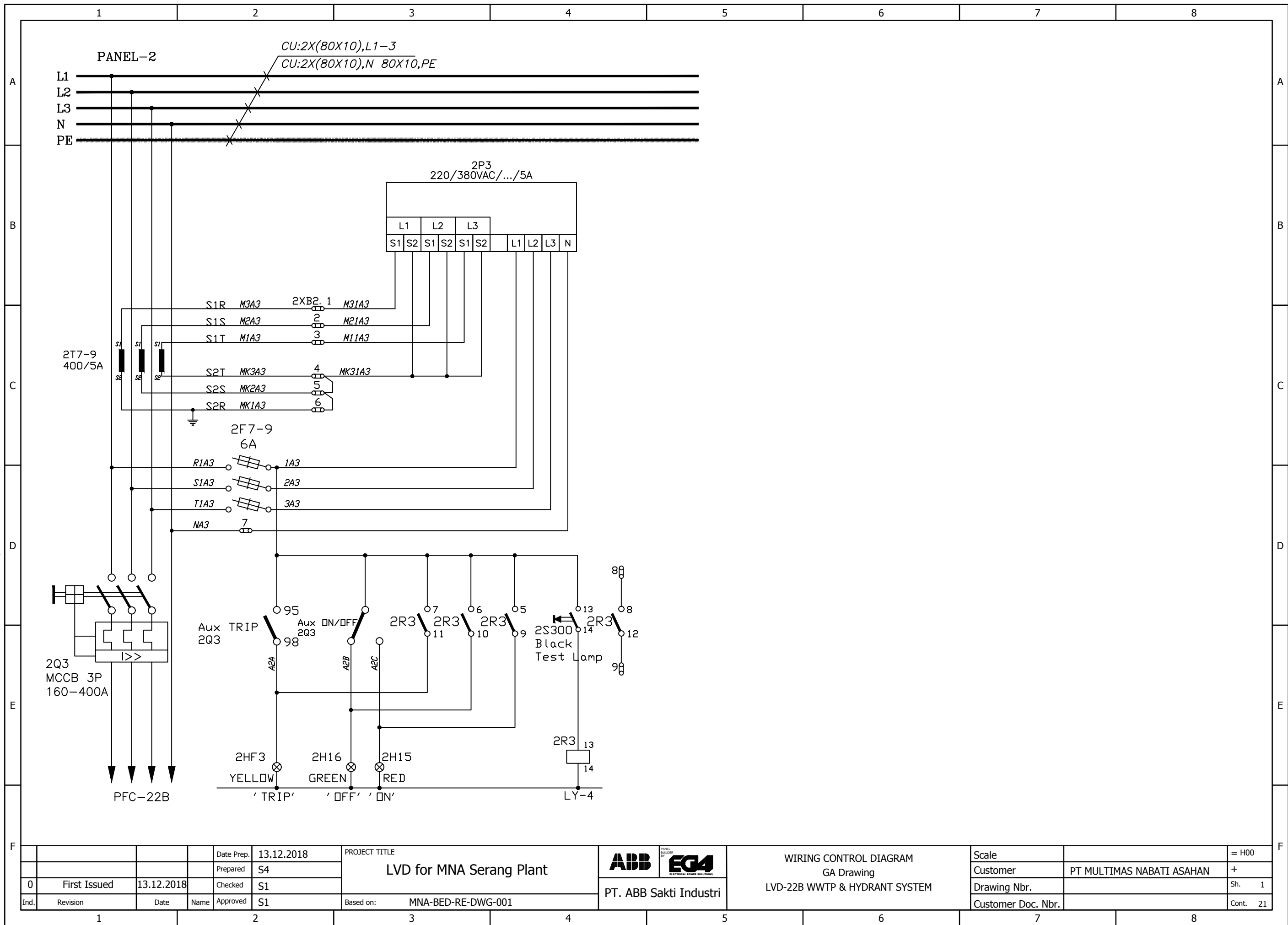
COUPLING

				Date Prep.	13.12.2018	PROJECT TITLE LVD for MNA Serang Plant	ABB PANEL BUILDER EG4 ELECTRICAL POWER SOLUTIONS	WIRING CONTROL DIAGRAM GA Drawing LVD-22B WWTP & HYDRANT SYSTEM	Scale		= H00
				Prepared	S4				Customer	PT MULTIMAS NABATI ASAHAN	+
0	First Issued	13.12.2018		Checked	S1				Drawing Nbr.		Sh. 1
Ind.	Revision	Date	Name	Approved	S1				Based on:	MNA-BED-RE-DWG-001	Customer Doc. Nbr.
1	2	3	4	5	6	7	8				

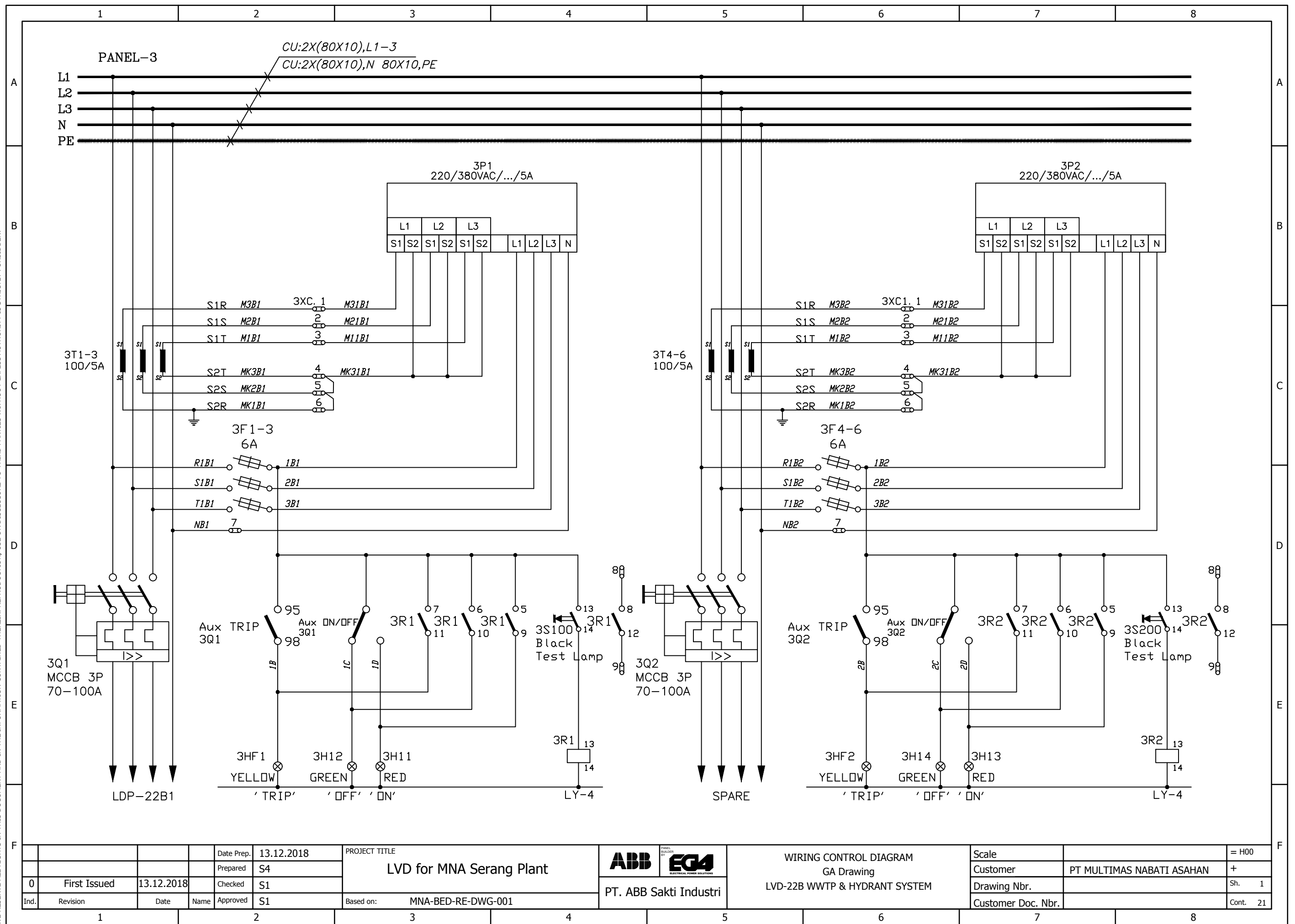
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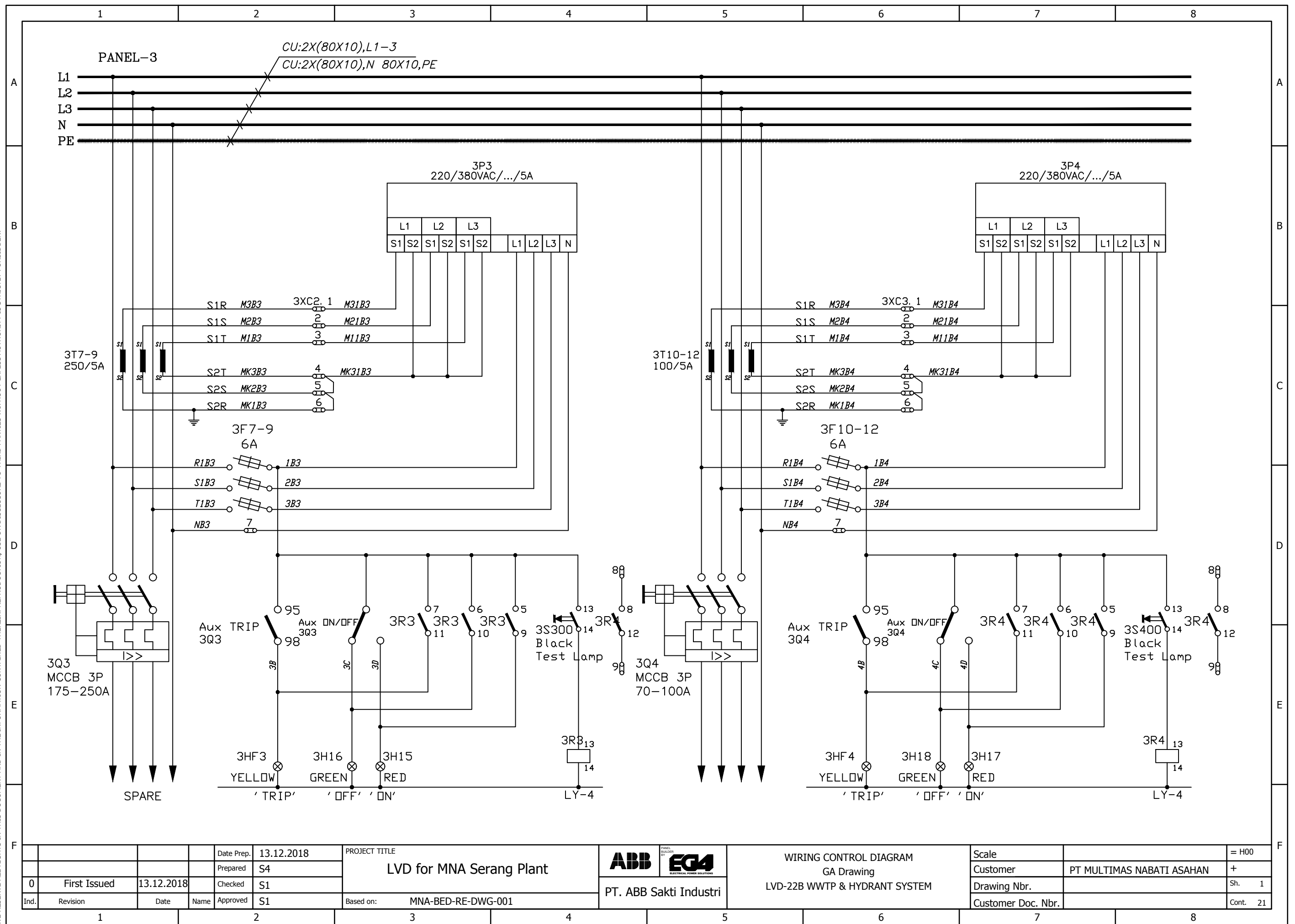
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1		2		3		4		5		6		7		8					
A		B		C		D		E		F		A		B					
N O		DESCRIPTION				TECHNICAL DATA				QTY		DESIGNATION							
		INCOMING																	
1		Fuse Carrier 6A				AXLE/Setara				7		1F11,1F1-6							
2		Pilot Lamp (LED) 220Vac/50Hz, Red				AXLE/Setara				2		1H1,11							
3		Pilot Lamp (LED) 220Vac/50Hz, Yellow				AXLE/Setara				2		1H2,HF1							
4		Pilot Lamp (LED) 220Vac/50Hz, Green				AXLE/Setara				2		1H3,H12							
5		DIGITAL POWER METER (*)				ABB, M2M ETHERNET				1		1P1							
6		CT 2000/5A CL.0.5 20VA				Howig, HG 82-2000				4		1T1-4							
7		ACB 3P 2000A, 66KA, DO, Ekip Touch LSIG (*)				ABB, E2.2N20 D/O				1		1Q0							
8		+ Shunt Trip Release (Y0) (*)				ABB, Y0				1		-							
9		+ Motor (M) + Closing Release (YC) For E2.2-E6.2 (*)				ABB, M+YC				1		-							
10		+ Padlocks For E2.2-E6.2 (*)				ABB,				1		-							
11		+ Ekip Com Protocol IEC 61850 (*)				ABB,				1		-							
12		Push Button On, 1NO, Green				Ega				1		S11							
13		Push Button Off, 1NO, Red				Ega				1		S12							
14		Control Relay 220Vac				Omron, LY-2				1		RF1							
15		Control Relay 220Vac				Omron, LY-4				2		R1-2							
16		Push Button Test Lamp , 1NO, Black				Ega				1		S100							
17																			
18		COUPLER																	
19		Fuse Carrier 6A				AXLE/Setara				2		1F8							
20		Pilot Lamp (LED) 220Vac/50Hz, Red				AXLE/Setara				1		1H14							
21		Pilot Lamp (LED) 220Vac/50Hz, Yellow				AXLE/Setara				1		1HF2							
22		Pilot Lamp (LED) 220Vac/50Hz, Green				AXLE/Setara				1		1H13							
23		ACB 4P 2000A, 66KA, FT, Ekip Dip LI (*)				ABB, E2.2N20 F/T				1		1Q01							
24		Push Button Test Lamp , 1NO, Black				Ega				1		S100							
25		Control Relay 220Vac				Omron, LY-4				1		R3							
26																			
F																F			
				Date Prep. 13.12.2018		PROJECT TITLE				ABB		EG4		COMPONENT LIST		Scale		= H00	
				Prepared S4		LVD for MNA Serang Plant								GA Drawing		Customer		PT MULTIMAS NABATI ASAHAN	
0		First Issued		13.12.2018		Checked S1						PT. ABB Sakti Industri		LVD-22B WWTP & HYDRANT SYSTEM		Drawing Nbr.		Sh. A+	
Ind.		Revision		Date		Name		Approved S1		Based on: MNA-BED-RE-DWG-001						Customer Doc. Nbr.		Cont. 21	
1		2		3		4		5		6		7		8					

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		1		2		3		4		5		6		7		8	
A	N O		DESCRIPTION				TECHNICAL DATA			QTY	DESIGNATION						A
			OUTGOING														
B	1		Fuse Carrier 6A				AXLE			21	2F1-9 ; 3F1-12						B
	2		Pilot Lamp (LED) 220Vac/50Hz, Red				AXLE			7	2H11,13,15 ; 3H11,13,15,17						
	3		Pilot Lamp (LED) 220Vac/50Hz, Yellow				AXLE			7	2HF1-3 ;3HF1-4						
	4		Pilot Lamp (LED) 220Vac/50Hz, Green				AXLE			7	2H12,14,16 ; 3H12,14,16,18						
	5		DIGITAL POWER METER (*)				ABB, M2M ETHERNET			7	2P1-3 ; 3P1-4						
	6		CT 800/5A CL.1 10VA				Howig, HG 50I-800			6	2T1-6						
	7		CT 400/5A CL.1 5VA				Howig, HG 40I-400			3	2T7-9						
	8		CT 250/5A CL.1 5VA				Howig, HG 30I-250			3	3T7-9						
C	9		CT 100/5A CL.1 2,5VA				Howig, HG 30I-100			9	3T1-6,10-12						C
	10		MCCB 3P 252-630A, 50KA (*)				ABB, T5S PR221DS-LS/I-630			2	2Q1-2						
	11		+ Auxiliary Contact For T5-T6 (*)				ABB, Aux-C 3Q 1SY			2	-						
	12		+ Extended Rotary Handles For T5 (*)				ABB, RHE			2	-						
	13		MCCB 3P 160-400A, 50KA (*)				ABB, T5S PR221DS-LS/I-400			1	2Q3						
	14		+ Auxiliary Contact For T5-T6 (*)				ABB, Aux-C 3Q 1SY			1	-						
D	15		+ Extended Rotary Handles For T5 (*)				ABB, RHE			1	-						D
	16		MCCB 3P 175-250A, 50KA (*)				ABB, XT3S TMD 250			1	3Q3						
	17		+ Auxiliary Contact For XT2/XT4 (*)				ABB, Aux-C 2Q 1SY			1	-						
	18		+ Extended Rotary Handles For XT2/XT4 (*)				ABB, RHE			1	-						
	19		MCCB 3P 70-100A, 50KA (*)				ABB, XT2S TMA-100			3	3Q1-Q2,Q4						
	20		+ Auxiliary Contact For XT2/XT4 (*)				ABB, Aux-C 2Q 1SY			3	-						
	21		+ Extended Rotary Handles For XT2/XT4 (*)				ABB, RHE			3	-						
E	22		Control Relay 220Vac				Omron, LY-4			7	2R1-3 ; 3R1-4						E
	23		Push Button Test Lamp , 1NO, Black				Ega			7	2S100-300 ; 3S100-400						
	24																
	25																
	26																
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